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# Toxic Nanoparticles Classification using Novel data augmentation and Artificial neural network

Susanta Mandal<sup>1</sup>, Sohini Paul<sup>1</sup>, Amit K. Chakraborty<sup>1</sup>, and Sayantari Ghosh<sup>1</sup>

National Institute of technology Durgapur, Department of Physics

**Abstract.** Nanoparticles(NPs) are progressively making their way into the daily lives of human beings through various mediums like medicines, consumer products, cosmetic products and many more. Their rising popularity is due to the fact that NPs show many promising properties which are not shown by the same material when used in bulk. However, on the other hand the small size of these NPs also give rise to toxic effects when they interact with the living tissues or the environment. In this paper, we design an in-silico data-driven methodology to classify nanoparticles according to their toxicity. With a extensive data collection from a large number of published manuscripts, nanoparticles with different sizes, shapes, materials, targeting various biological targets in different experimental conditions were considered. With a dataset created and enhanced, a deep model using Artificial Neural Network has been developed to detect the toxicity of nanoparticles. The model, even with a moderate-size dataset, trains quite well resulting into a performance accuracy of more than 93%.

**Keywords:** Nanotoxicity, Deep learning, New dataset, Classification, Feature identification.

## 1 Introduction

Nanotechnology is rapidly making its place in the human life through various commercial sources. Nanoparticles show more toxicity compared to larger particles owing to their increased freedom of movement which is much higher compared to bulkier molecules. The absorption of the nanoparticles through the skin is much easier because the nanoparticles can smoothly penetrate the surface layer of skin from the cosmetic products due to diffusion . Nanoparticles like TiO<sub>2</sub>, Au, Ag, and polymers are insoluble as they are quite stable. As a result, they directly cause harm once they enter the body [1].

Nanomaterials are abundantly incorporated into the cosmetic products. Preservation of youthfulness has always been in demand and the antioxidant properties of nanoparticles aid in this area [2]. Nanoparticles though were considered to be biologically harmless initially, recent studies [3] have revealed otherwise. The incorporation of Ag nanoparticles into various products have increased due to its antibacterial and antifungal properties. Silver is extensively used as an antibacterial agent. But researches prove that it can be toxic to humans and also animal cells [4]. Application of nanomedicine in early diagnosis and hence effective treatment is extremely vital. Amongst different nanoparticles, superparamagnetic iron oxide nanoparticles (SPIONs) are predominantly effective for magnetic resonance imaging (MRI). The paramagnetic properties of SPIONs and the fact that their toxicity levels are low i.e., below 100 mg/mL make them excellent candidates. Though SPIONs have wide applications in various fields, these are known to be cytotoxic, and hence can damage the functions of nucleus, DNA and mitochondria [5].

While more cautious handling and manipulation of engineered nanoparticles, increased validation and standardization of nanoparticle toxicity tests, is essential to encourage safety while incorporating nanoparticles in daily life; computational tools can become extremely handy to predict properties of nanoparticles beforehand. Deep learning has the ability to analyse large datasets through several layers of interconnected neurons and also update itself automatically in order to provide the most accurate results. The different components of deep networks provide different levels of accuracy and are used for applications in different types of data processing. In this paper, the aim is to apply deep learning-based tools to explore and predict nanotoxicity effectively.

It is interesting to note that compared to substitutional data analysis approaches, deep learning methods are not in requirement of user pre-defined data descriptors. The training procedure enables the model to handle a very complex feature space automatically. Thus, several problems with the existing methodologies could be overcome using deep learning methods. Moreover, these training algorithms are capable of processing big datasets (in the order of thousands) in relatively short computational times.

Artificial neural network-based learning is making difference in the day to day life. Artificial Intelligence aims to replicate the human brain and the most challenging task in order to do this is high level data processing. The rising growth and application of artificial intelligence and machine learning, and availability of computers capable of handling large-scale data, is making the effect of these machine intelligence tools

indispensable in fields like, image processing [6], computer vision [7], bioinformatics [8], drug discovery [9] and natural language processing [10]. In this paper, a deep framework has been designed which is capable of predicting toxicity of unknown nanoparticles based on available experimental data training. The major goals of this paper are to:

- Explore nanotoxicity with focus on types and specificities of toxic effects.
- Create and enhance existing datasets for nanotoxicity detection.
- Propose an innovative deep neural network to handle numerical as well as categorical data.
- Create a holistic *in-silico* methodology for nanotoxicity prediction.

## 2 Materials and Methods

### 2.1 Dataset

The data that was used for training and validation of the model was compiled from numerous research papers. For the computational experiment, the compilation of the dataset was done by a detailed review of experiments carried out by several other researchers. Finally, a dataset containing information of 460 nanoparticles was compiled. 260 data points were collected from the research paper of Concu et al. [11]. The remaining 200 data was collected after rigorous review of a number of research papers [12, 13, 14, 15, 16, 17, 18]. In total 178 papers were reviewed and 100 research papers dealing with experimental observations of toxicity of nanoparticles were chosen. Each paper was methodically reviewed for all the parameters. Detailed analysis was done to obtain a dataset rich in both toxic as well as non-toxic data points. This was necessary for obtaining a highly accurate model. The various parameters that were considered for these nanoparticles were discussed in detail below. To preserve parity, we have used similar indicating symbols for these features following Concu et.al.

- Type ( $m$ ): The different types of nanoparticles in this paper includes metals (Au, Ag, Pt, Pd, etc.), metal oxides (ZnO, CuO, TiO<sub>2</sub>, Fe<sub>2</sub>O<sub>3</sub>, etc.), semiconductors (Si) and their oxides, etc. A total of 32 types of nanoparticles have been used.
- Size ( $r$ ): A wide range of size was considered for the different nanoparticles. It ranged from approximately 1 nm to 300 nm.
- Shape ( $s$ ): The different shapes of the nanoparticles included spherical, rod-shaped, elliptical, pyramidal etc. The total number of different shapes considered was 10.
- Toxicity Measures ( $tx$ ): The toxicity measures EC<sub>50</sub>, IC<sub>50</sub>, LC<sub>50</sub>, TC<sub>50</sub> and CC<sub>50</sub> were considered.
- Toxicity measure values ( $tx_v$ ): The reported values of respective toxicity measures, as experimentally detected.
- Biological End-points ( $bio$ ): The targeted biological endpoints included algae, crustaceans, fungi, fish and many other cell lines. The total number of different end-points was 83.
- Surface Coating ( $S_{coat}$ ): Some of the nanoparticles were coated with some chemical while some were left uncoated. 25 different types of chemical was used for coating the nanoparticles.
- Conditions for Size determination ( $d_c$ ): The sizes of the nanoparticles were determined under various conditions, for example in dry form, in various solutions (RPMI, DMEM, H<sub>2</sub>O, etc.). The total number of conditions evaluated was 10.
- Assay Time ( $t$ ): The assay time varied between 0.5 hrs and 360 hrs.

The atomic volume (V), polarizability (P) and electronegativity (E) of the nanoparticles was also considered. It was made sure that no data for these features were missing in the research studies that were being considered.

### 2.2 Data Encoding

The data that was obtained after compilation contained numerical data for some of the categories, like size, volume, polarizability etc., while others had categorical data e.g. shape, the toxicity measures used ( $m_e$ ), biological target cells ( $b_t$ ) etc. The model cannot work with categorical data. Hence encoding of categorical data into numerical data was necessary. In this research the method of Label encoding was found to be most suitable for the prepared dataset.

## 2.3 Data Labelling

The dataset containing the data of 460 nanoparticles was divided into two groups, 'nontoxic' and 'toxic' based on the threshold mentioned in Table 1. Nanoparticles were assumed nontoxic i.e.  $[L_i = 1]$  if their toxicity values lie in any one of the ranges mentioned in Table 1. If not, then nanoparticles were assumed as toxic  $[L_i = -1]$ . Here,  $L_i \in L$  where  $L = \{-1, 1\}$  is a set of binary category variable that has been used as a label for the purpose of classifying the nanoparticles into two groups according to the criteria of toxicity. The two key reasons which are responsible for taking on such criteria are listed as follows.

Table 1: Table showing various toxicity terms along with Threshold Values

| Toxicity Terms Threshold Values ( $\mu M$ ) |               |
|---------------------------------------------|---------------|
| CC50                                        | $\geq 188.57$ |
| EC50                                        | $\geq 168.46$ |
| IC50                                        | $\geq 245.68$ |
| TC50                                        | $\geq 126.52$ |
| LC50                                        | $\geq 250$    |

- The nanoparticle concentrations that are usually used are much higher than  $\mu M$ . Owing to this fact, the thresholds that have been selected are in ranges high enough to ensure that the nanoparticle that has been predicted as nontoxic was correct and it really did not have any toxic reactions.
- The thresholds that have been selected will lead to a sensible proportionate ratio between nontoxic and toxic nanoparticles in the collected dataset.

Also, the creation a well-balanced dataset is very critical for designing a successful deep model. The difficulty of class imbalance becomes more prominent when the availability of one type of data (i.e., either toxic or nontoxic) is greater than the other. Training a deep network with class imbalanced data creates unstable output that is inherently biased towards the larger available class. Thus, extreme care was taken while creating the dataset. After labelling of the dataset there was a total of 460 data points, in a 12-dimensional feature space where 238 nontoxic and 222 toxic samples were present. Thus, the dataset did not suffer from the problem of class imbalance.

## 2.4 Model elements

The model described in this consist of Artificial Neural Network. The elements of the Neural Network are

- Fully Connected or *Dense* Layers: In neural networks, fully connected layers are those layers in which each neuron is connected to every neuron in the next layer.
- Activation Function: The activation function is most important for training the model as it is responsible for updation of the weights and biases related with each neuron. There are various types of activation functions. In this model, ReLu or Rectified Linear Unit activation function and Sigmoid function have been used.

$$\theta_{ReLU}(x) = \max(x, 0) \quad (1)$$

ReLU activation function has resemblance with the neurons of the human brain. It returns a 0 when faced with a negative value and the value remains unchanged if it faces a positive value. This non-linear activation function is usually applied in the hidden layers of the Neural network [19].

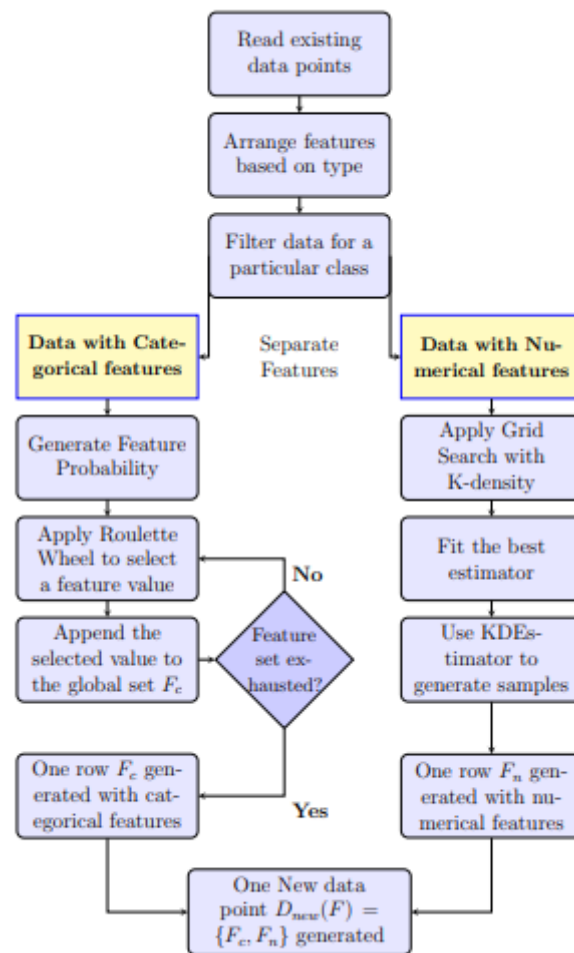


Fig. 1: Flow of a data point generation specific to a class.

Sigmoid activation function produces an 'S' shaped plot on the graph. It's equation is given by

$$\theta_{Sig}(x) = \frac{1}{1 + e^{-x}} \quad (2)$$

It is a non-linear function ranging from 0 to 1. Sigmoid function is generally implemented in the output layer if a binary classification model is being designed. Here the obtained result will be a binary value i.e. either 0 or 1. The result will be predicted as 1 if the value is greater than a mid-value of 0.5. Else it will be 0.

- Batch Normalization: Batch Normalization is a transformation technique used for normalizing the inputs in the hidden layers. Constantly updating and normalizing the inputs enhances training accuracy and the stability of the neural network. This technique transforms the input of each layer by normalizing the output obtained from the previous activation layer. The batch mean is subtracted from each output and then it is divided by the batch standard deviation. Two trainable parameters; standard deviation parameter  $\beta$  and mean parameter  $\alpha$  are then added. After normalization, the output is multiplied by  $\beta$  and then  $\alpha$  is added to it. The inputs in the batch normalization operation are the values of  $x$  in a batch  $[B = (x_{1,2}, \dots, x_n)]$  formed by the outputs from the previous layer.  $m_B$  and  $\sigma_B$  are the batch mean and batch standard deviation respectively [20]. The initial normalization is computed according to equation 3.

$$x'_i = \frac{x_i - m_B}{\sqrt{\sigma_B^2 + \phi}} \quad (3)$$

After the normalization, the trainable parameters are incorporated according to equation 4.

$$y'_i = \beta x'_i + \alpha \quad (4)$$

- Dropout: In neural networks overfitting may occur when too much data is fed to a neuron while training the model. To avoid this Dropout technique is utilised in the network. In this technique, neurons are selected randomly and they are dropped during training. As a result, the neuron will not convey any data to the next layer and also the weights associated with the neuron will not be updated through back propagation.

## 2.5 Model Architecture

For successful design of an automated system, capable of classification of toxic and nontoxic nanoparticles, we chose ANN network. In our proposed network we have utilized the above mentioned elements in different layers of the network. We have sequentially added three Dense layers with dimension of 60, 20 and 1 respectively. Intermediately, after the first Dense layer we have performed Batch Normalization. Dropout (with dropout value 0.3) has been inserted which generalized the overall network, otherwise the network will be over-fitted with the vast number of learnable parameters provided in the Dense layers. Three activation functions have been used, i.e. two ReLu and sigmoid at the last. The major advantage of our proposed network over the MLP Classifier is the use of BatchNormalization and the Dropout which are not there in the basic MLP. We have compared different indexes of the proposed networks with the K-Nearest Neighbour, SVM, Random Forest, Decision Tree, and MLP for establishing the superiority. The total number of parameters in the proposed model is 4,913, out of which number of trainable parameters is 4,593 and number of non-trainable parameters is 320.

## 3 Result and Discussion

### 3.1 Experimental Set up

The experiments were performed in Google Colaboratory platform with the data file of features compiled into a .csv file. The programming language used was Python.

### 3.2 Authenticity of generation method

Before starting the training of the model with the created dataset, it was necessary to ensure the authenticity of the generation method. To achieve that we study the feature space for feature correlations for both real and generated dataset.

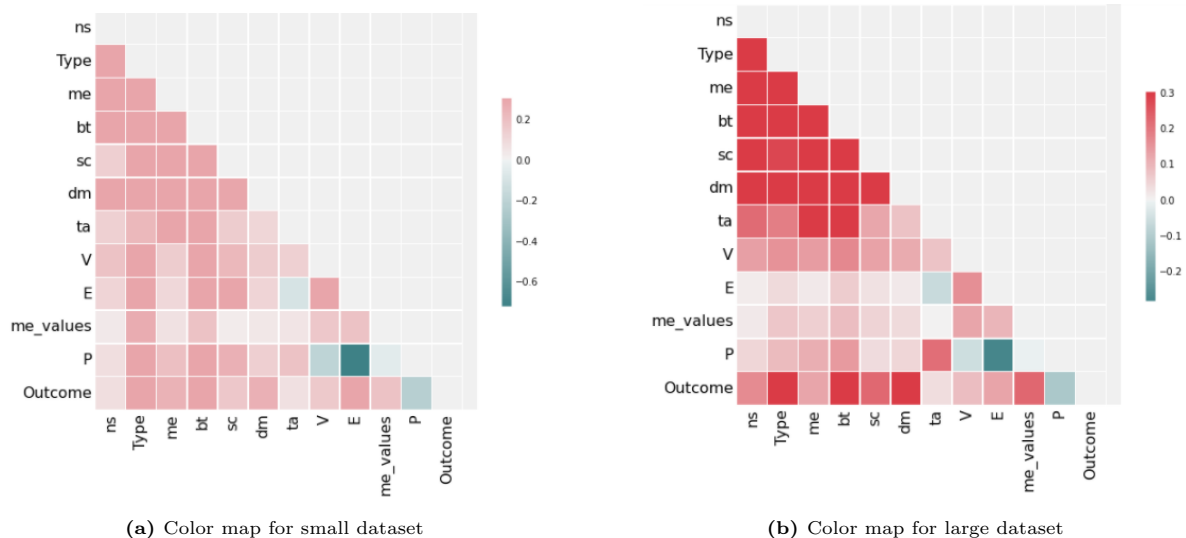


Fig. 2: Feature Correlation color map matching for small dataset and generated large scale dataset

### 3.3 Classification Experiments & Results

The model was trained with a random split of 0.5 for the dataset i.e., half of the random data were used for training the model, and half were used for validation of the learning of the model. The model was trained for 140 epochs, achieving a validation accuracy of 93.04%. This signifies that for several nanoparticles which were never shown to the model at the time of training, the machine is capable of predicting in toxic behaviours with 93% certainty.

**Accuracy vs. epochs** The accuracy vs. epochs plot shows the initial accuracy of the model to be at about 45-50%. Initially both the training and validation accuracy was same. As the number of epochs i.e. the number of cycles of the model increased, the training accuracy as well as the validation accuracy increased gradually. The training and validation accuracy remained at a comparable level with each other up to 115-118 epochs, after which the validation accuracy became more. The validation accuracy reached more than 90% while the training accuracy remained at 70%.

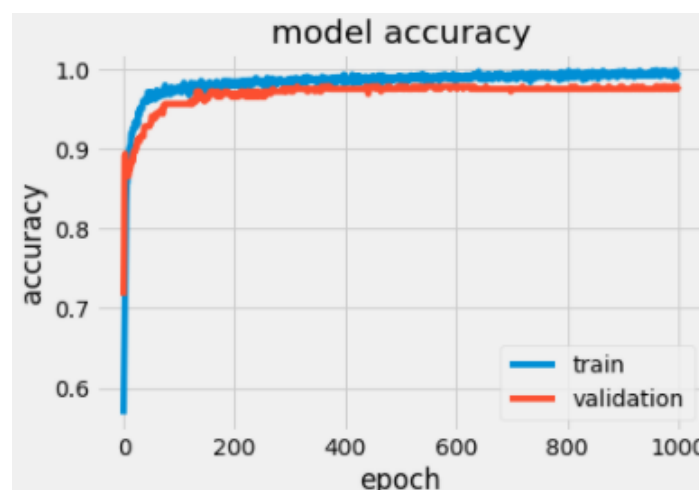


Fig. 3: Accuracy (y-axis) vs epochs (x-axis) for a sample (randomly chosen) K-Fold set

**Loss vs. epochs** The loss vs. epochs plot shows the loss minimization while training and validation of the data. It was seen that the initial loss was around 80% while training and around 70% while validating. The loss decreased uniformly and the final loss percentage for both the training and validation set was

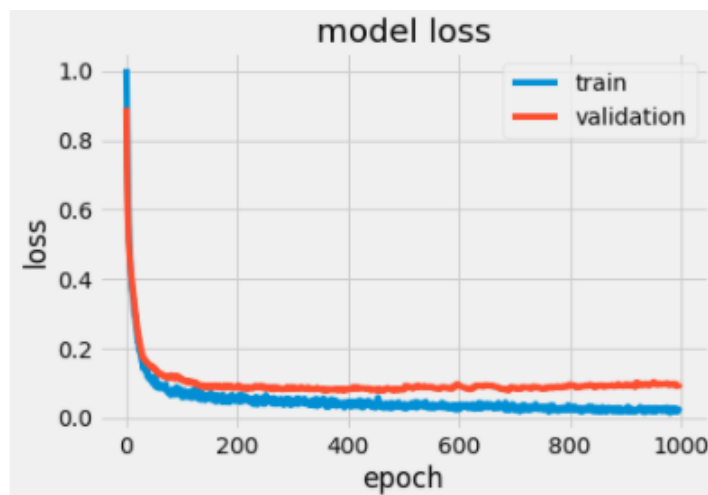


Fig. 4: Loss (y-axis) vs epochs (x- axis) for a sample (randomly chosen) K-Fold set

around 55-60%. The increase in validation accuracy with a decrease in validation loss shows that the model that has been built is a good model without overfitting or underfitting.

**Confusion matrix** In Machine learning, confusion matrix is an important technique which is used to measure the performance of the model. It is the matrix representation of the predicted values and actual values.

- True Positives (TP) - Correctly predicted positive values.
- True Negatives (TN) - Correctly predicted negative values.
- False Positives (FP) - When predicted class is positive but actual class is negative.
- False Negatives (FN) - When predicted class is negative but actual class is positive.

$$\text{Precision}(Pr) = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalsePositive}}$$

$$\text{Recall}(Re) = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalseNegative}}$$

$$F1Score = \frac{2 * (Pr * Re)}{Re + Pr}$$

Figure 5 shows the confusion matrix of our model. The matrix shows that it has been able to predict 114 nontoxic nanoparticles as nontoxic, 100 toxic nanoparticles as toxic, 4 toxic nanoparticles as nontoxic and 12 nontoxic nanoparticles as toxic nanoparticle. Hence, it can be seen that the model has been able to predict most of the values correctly and it has a high accuracy.

|                 |          | Actual Label |       |
|-----------------|----------|--------------|-------|
|                 |          | Nontoxic     | Toxic |
| Predicted Label | Nontoxic | 114          | 4     |
|                 | Toxic    | 12           | 100   |

Fig. 5: Confusion Matrix of the prepared model

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**Precision, Recall, F1-score and Support** There are some standard statistical measures for estimating performances of a neural network model. These are Precision, Recall, F1-score and Support. The measures obtained for the model in this paper is shown in Table 2 and Table 3.

Table 2: Model Report showing values of precision, recall, f1-score and support: Training Set

Accuracy:0.9186

Classification Report

|         | precision | recall | f1-score | Support |
|---------|-----------|--------|----------|---------|
| Class-1 | 0.91      | 0.76   | 0.85     | 154     |
| Class-2 | 0.94      | 0.90   | 0.81     | 168     |

Table 3: Model Report showing values of precision, recall, f1-score and support: Test Set

Accuracy:93.33%

Classification Report

|         | precision | recall | f1-score | Support |
|---------|-----------|--------|----------|---------|
| Class-1 | 0.96      | 0.89   | 0.92     | 68      |
| Class-2 | 0.91      | 0.87   | 0.84     | 70      |

## 4 Conclusion

In this paper, we have developed a computational model using Deep Neural Network for the prediction of toxicity of nanomaterials. Looking at this model, it can be understood how much easier it is to predict the toxicity compared to experimental approaches. The right tuning of the neural network, by adding different numbers of dense layers of varied density and the use of the right activation function enabled us to obtain a really high percentage of accuracy. This shows that the model we have developed is a very good model which can be used for prediction of toxicity. The model that we have developed can be used for prediction of toxicities in various areas of science, technology and research, where nanoparticles get used frequently. Using this model for additional research could also aid in the creation of better models with higher accuracy than what we now have. The model will be trained more effectively and accurately with a larger dataset than the one used in this research. Computational approaches are a great alternative when experimental would not produce the best results or could be expensive. Furthermore, deep models can learn on their own from provided data, eliminating the need for tedious model training. This will be addressed in a future work.

## Supplementary Material

The entire list of research works which have been used for dataset preparation are listed in the given link<sup>1</sup>.

<sup>1</sup> <https://docs.google.com/spreadsheets/d/1NkbE05YAy82LONM8vjew3PAGYZJg1Nfc/edit?usp=drive.link&ouid=103931989926520852998&rtpof=true&sd=true>

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## Bibliography

- [1] Juergen Lademann, Hans-Juergen Weigmann, Christiane Rickmeyer, Hans Barthelmes, Hans Schaefer, Gerhard Mueller, and Wolfram Sterry. Penetration of titanium dioxide microparticles in a sun-screen formulation into the horny layer and the follicular orifice. *Skin pharmacology and physiology*, 12(5):247–256, 1999.
- [2] Yan Zhang. *Blends of cyclic poly (butylene terephthalate)/multiwalled carbon nanotube nanocomposites prepared by in situ polymerization*. PhD thesis, University of Birmingham, 2015.
- [3] B Rehn, F Seiler, S Rehn, J Bruch, and M Maier. Investigations on the inflammatory and genotoxic lung effects of two types of titanium dioxide: untreated and surface treated. *Toxicology and applied pharmacology*, 189(2):84–95, 2003.
- [4] Albanelly Soto-Quintero, Nekane Guarrotxena, Olga García, and Isabel Quijada-Garrido. Curcumin to promote the synthesis of silver nps and their self-assembly with a thermoresponsive polymer in core-shell nanohybrids. *Scientific reports*, 9(1):1–14, 2019.
- [5] Neenu Singh, Gareth JS Jenkins, Romisa Asadi, and Shareen H Doak. Potential toxicity of superparamagnetic iron oxide nanoparticles (spion). *Nano reviews*, 1(1):5358, 2010.
- [6] Olivier Lézoray, Christophe Charrier, Hubert Cardot, and Sébastien Lefèvre. Machine learning in image processing, 2008.
- [7] Pedro J Navarro, Fernando Pérez, Julia Weiss, and Marcos Egea-Cortines. Machine learning and computer vision system for phenotype data acquisition and analysis in plants. *Sensors*, 16(5):641, 2016.
- [8] Yan-Qing Zhang and Jagath Chandana Rajapakse. *Machine learning in bioinformatics*, volume 4. Wiley Online Library, 2009.
- [9] Antonio Lavecchia. Machine-learning approaches in drug discovery: methods and applications. *Drug discovery today*, 20(3):318–331, 2015.
- [10] Arjan S Gosal, Ilse R Geijzendorffer, Tomáš Václavík, Brigitte Poulin, and Guy Ziv. Using social media, machine learning and natural language processing to map multiple recreational beneficiaries. *Ecosystem Services*, 38:100958, 2019.
- [11] Riccardo Concu, Valeria V Kleandrova, Alejandro Speck-Planche, and M Natália DS Cordeiro. Probing the toxicity of nanoparticles: a unified in silico machine learning model based on perturbation theory. *Nanotoxicology*, 11(7):891–906, 2017.
- [12] Hainan Sun, Yin Liu, Xue Bai, Xiaofei Zhou, Hongyu Zhou, Sijin Liu, and Bing Yan. Induction of oxidative stress and sensitization of cancer cells to paclitaxel by gold nanoparticles with different charge densities and hydrophobicities. *Journal of Materials Chemistry B*, 6(11):1633–1639, 2018.
- [13] Xue Bai, Shenqing Wang, Xiliang Yan, Hongyu Zhou, Jinhua Zhan, Sijin Liu, Virender K Sharma, Guibin Jiang, Hao Zhu, and Bing Yan. Regulation of cell uptake and cytotoxicity by nanoparticle core under the controlled shape, size, and surface chemistries. *ACS nano*, 14(1):289–302, 2019.
- [14] Shin Woong Kim, Sun-Hwa Nam, and Youn-Joo An. Interaction of silver nanoparticles with biological surfaces of caenorhabditis elegans. *Ecotoxicology and environmental safety*, 77:64–70, 2012.
- [15] Joanna M Seiffert, Marc-Olivier Baradez, Volker Nischwitz, Tamara Lekishvili, Heidi Goenaga-Infante, and Damian Marshall. Dynamic monitoring of metal oxide nanoparticle toxicity by label free impedance sensing. *Chemical research in toxicology*, 25(1):140–152, 2012.
- [16] Jean-Pascal Piret, Diane Jacques, Jean-Nicolas Audinot, Jorge Mejia, Emmanuelle Boilan, Florence Noël, Maude Fransolet, Catherine Demazy, Stéphane Lucas, Christelle Saout, et al. Copper (ii) oxide nanoparticles penetrate into hepg2 cells, exert cytotoxicity via oxidative stress and induce pro-inflammatory response. *Nanoscale*, 4(22):7168–7184, 2012.
- [17] Zhuang Wang, Jingwen Chen, Xuehua Li, Jianping Shao, and Willie JGM Peijnenburg. Aquatic toxicity of nanosilver colloids to different trophic organisms: contributions of particles and free silver ion. *Environmental Toxicology and Chemistry*, 31(10):2408–2413, 2012.
- [18] Sarah M Hoheisel, Steve Diamond, and David Mount. Comparison of nanosilver and ionic silver toxicity in daphnia magna and pimephales promelas. *Environmental toxicology and chemistry*, 31(11):2557–2563, 2012.
- [19] Shan Sung Liew, Mohamed Khalil-Hani, and Rabia Bakhteri. Bounded activation functions for enhanced training stability of deep neural networks on visual pattern recognition problems. *Neuro-computing*, 216:718–734, 2016.
- [20] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.